Mental Health Predictive Analytics Project

ISOM 835 – Predictive Analytics and Machine Learning

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Introduction & Dataset Description

For this project, I chose to work with a mental health dataset from Kaggle that focuses on how lifestyle habits and daily behaviors relate to stress levels. Mental health is something that has become more widely discussed in recent years, especially with the increase in screen time, social media usage, and overall busy lifestyles. Because of that, I thought this dataset would be both interesting and relevant to something that affects people in a real way.

The dataset includes 2,000 observations and contains a variety of variables related to an individual's lifestyle and background. Some of the main variables include age, gender, occupation, daily screen time, social media usage, sleep hours, work or study hours, physical activity, caffeine intake, and other behavioral factors. It also includes variables related to mental health such as anxiety, depression, and burnout, which help provide additional context.

The main goal of this project is to determine whether stress level can be predicted based on these types of variables. In addition to that, I wanted to identify which factors seem to have the biggest influence on stress. This type of analysis could be useful in real-world situations, such as workplace wellness programs, mental health awareness tools, or even personal habit tracking apps.

At the same time, it is important to recognize that mental health is complex and not something that can be perfectly captured through data alone. This idea became more apparent as the project progressed.

Exploratory Data Analysis (EDA)

Before building any models, I started by exploring the dataset to better understand its structure and the distributions of the variables. This step is important because it helps identify patterns, potential issues, and relationships that might influence the modeling process.

The dataset includes both numerical and categorical variables. Numerical variables such as age, daily screen time, sleep hours, and work/study hours showed a wide range of values. This variation is useful because it provides enough diversity for the models to learn from.

When looking at age, the dataset includes individuals across a broad range, which helps avoid bias toward a specific group. The gender distribution also appeared fairly balanced, which is important when considering fairness in modeling.

Daily screen time and social media usage stood out as particularly interesting variables. Some individuals reported very low usage, while others reported significantly higher values. This suggests that technology use varies widely and could potentially play a role in stress levels.

Sleep hours also varied across individuals. Since sleep is commonly linked to mental health, this was one of the variables that seemed likely to be important in predicting stress. Similarly, work or study hours showed noticeable differences between individuals, which could also contribute to stress.

The correlation heatmap provided a general overview of relationships between variables. While some variables showed moderate relationships, there were no extremely strong correlations. This suggests that stress is not driven by a single factor, but rather by a combination of influences.

Another important observation from the EDA was the variability within similar groups. For example, individuals with similar ages or occupations still showed different stress levels. This highlights the complexity of mental health and suggests that it may be difficult to predict accurately using only these variables.

Overall, the EDA helped set expectations for the modeling phase. It showed that while there are some patterns in the data, stress is likely influenced by multiple overlapping factors rather than one clear predictor.

Data Cleaning and Preprocessing

After exploring the dataset, the next step was to prepare it for modeling. This involved cleaning the data and making sure it was in a format that machine learning models could use.

The first step was to remove the Person_ID column. This column serves only as an identifier and does not provide any meaningful information for predicting stress levels. Keeping it in the dataset could actually introduce noise into the model.

Next, I handled the categorical variables. Since machine learning models require numerical input, I used label encoding to convert variables such as gender and occupation into numeric form. This allows the models to process these variables while still preserving their categories.

It is worth noting that label encoding can sometimes introduce limitations, as it assigns numerical values to categories that do not necessarily have a meaningful order. However, for the models used in this project, this approach was sufficient.

I also checked for missing values in the dataset. Fortunately, there were no major missing values that needed to be addressed, which made the preprocessing process more straightforward.

After cleaning the data, I split it into training and testing sets using an 80/20 split. This means that 80% of the data was used to train the models, while the remaining 20% was used to evaluate their performance. This is a standard approach that helps ensure the model is tested on data it has not seen before.

For the Logistic Regression model, I applied feature scaling using a standard scaler. This step ensures that all variables are on a similar scale, which is important for models that are sensitive to differences in magnitude. Without scaling, variables with larger values could have a disproportionate impact on the model.

Overall, the preprocessing steps were relatively straightforward, but they were essential for preparing the data and ensuring that the models could run correctly.

Business Questions

This project was guided by the following business-related questions:

1. How do lifestyle factors such as sleep, screen time, and physical activity impact stress levels?
2. Can stress levels be accurately predicted using behavioral and demographic data?
3. Which variables are the strongest predictors of stress level?

These questions helped focus the analysis and provided a clear purpose for the modeling process. Rather than simply building models, the goal was to gain insight into how different factors relate to stress.

Predictive Modeling

To address these questions, I used two different machine learning models: Logistic Regression and Random Forest.

Logistic Regression was chosen as a baseline model because it is simple and easy to interpret. It assumes a linear relationship between the input variables and the output, which can be useful for understanding general trends.

Random Forest, on the other hand, is a more advanced model that can capture non-linear relationships. It works by combining multiple decision trees, which allows it to identify more complex patterns in the data.

After training both models, I evaluated their performance using accuracy and F1 score.

The results were as follows:

- Logistic Regression Accuracy: 0.125
- Logistic Regression F1 Score: 0.116
- Random Forest Accuracy: 0.130
- Random Forest F1 Score: 0.123

The Random Forest model performed slightly better than Logistic Regression, but overall, both models had relatively low performance.

This suggests that predicting stress level using this dataset is challenging. It may indicate that important variables are missing or that stress is influenced by factors that are not easily captured in this type of data.

Insights and Interpretation

Although the models did not perform particularly well, the results still provide useful insights.

The low accuracy suggests that stress is not easily predictable using basic lifestyle data alone. This reinforces the idea that mental health is complex and influenced by many different factors, including psychological and environmental elements that may not be included in the dataset.

Despite this, the feature importance results from the Random Forest model offer valuable information. The most important variables included daily screen time, work/study hours, social media usage, sleep hours, and age.

These findings suggest that how individuals spend their time plays a significant role in stress levels. For example, high screen time and heavy social media usage may contribute to increased stress, while adequate sleep may help reduce it.

It is also important to consider that these factors likely interact with each other. For example, someone who has high screen time, low sleep, and long work hours may experience higher stress than someone who only has one of those factors.

Even though the models were not highly accurate, they still help identify patterns and highlight which areas may be most relevant when thinking about stress.

Ethics and Interpretability

There are several ethical considerations associated with this type of analysis.

First, mental health data is highly sensitive, and any predictions made by a model should be handled carefully. The relatively low accuracy of the models also raises concerns about reliability. Using such models to make real-world decisions could lead to incorrect conclusions about individuals.

Another important consideration is bias. The dataset may not fully represent all populations, which could result in models that perform better for some groups than others. This could lead to unfair or misleading outcomes.

Interpretability is also important. Stakeholders should be able to understand how predictions are being made, rather than relying on a model that functions as a “black box.”

Overall, models like this should be used as tools to support understanding, rather than as definitive decision-making systems.

Conclusion

This project explored how predictive analytics can be applied to a mental health dataset to better understand stress levels.

While the models did not achieve high accuracy, the analysis still provided meaningful insights. It highlighted the importance of lifestyle factors such as screen time, sleep, and workload, and demonstrated how these variables may relate to stress.

The results also emphasized the limitations of using data to predict mental health outcomes. Stress is influenced by many factors, and not all of them can be captured in a dataset.

Overall, this project provided a valuable learning experience and reinforced the importance of both technical skills and critical thinking when working with data.

APPENDIX:

Full Code: https://colab.research.google.com/drive/19LEXH_NkcNRi6-04kGabZAMZzmyn9Nk9?usp=sharing

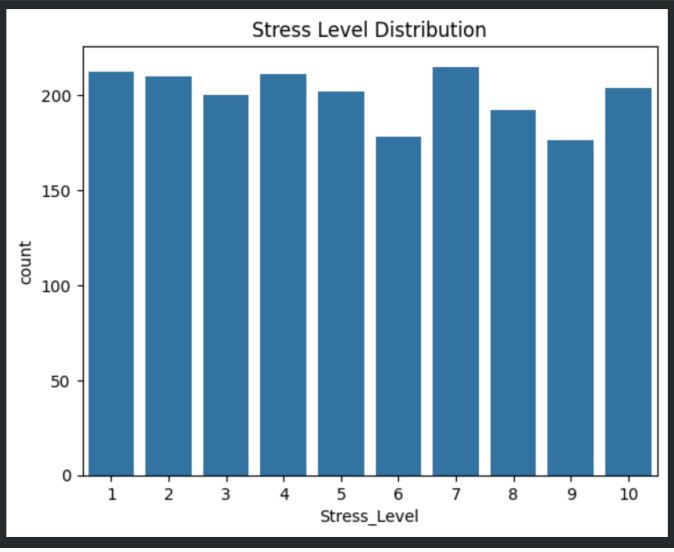
[33]
0s

```
feature_importance
```

	Feature	Importance
3	Daily_Screen_Time	0.120466
7	Work_Study_Hours	0.119641
4	Social_Media_Usage	0.116473
6	Sleep_Hours	0.113806
0	Age	0.109212
9	Social_Interaction_Score	0.086977
10	Caffeine_Intake	0.065243
2	Occupation	0.046643
8	Physical_Activity	0.045653
1	Gender	0.030139
12	Alcohol	0.029411
5	Night_Usage	0.028678
11	Smoking	0.028578
14	Anxiety	0.020318
15	Burnout	0.020056
13	Depression	0.018706

[18]
0s

```
sns.countplot(x='Stress_Level', data=df)
plt.title("Stress Level Distribution")
plt.show()
```



[32]
0s

```
results
```

	Model	Accuracy	F1 Score
0	Logistic Regression	0.125	0.115757
1	Random Forest	0.130	0.122998

Next steps: Generate code with results | New interactive sheet